

AKSHAY PAGHADAR

Electronics Engineer | Embedded Systems & Robotics | IoT Design

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Professional Summary

Final-year Electronics Engineering student with hands-on experience designing embedded, IoT, and robotic systems from concept to field deployment. National award winner (Robofest 4.0) and two-time patent holder for an autonomous underwater vehicle that placed 6th at SAUVC 2025, Singapore, against 20+ international teams. Skilled in PCB design, firmware development, and LoRa wireless communication, with strong project leadership on low-cost solutions for agriculture, safety, and marine robotics. Seeking embedded systems, firmware, or robotics engineering roles, or a Master's program, in Europe.

Education

Gujarat Technological University – Vallabh Vidyanagar, India

B.Tech in Electronics Engineering

2022 – 2026

- CGPA: 7.68 / 10.0

Technical Skills

Programming	C, C++, Python (Basic)
Embedded & Hardware	Embedded Systems Design, Microcontroller-Based Systems, PCB Design, Sensor Integration
Wireless & Communication	LoRa Wireless Communication, IoT Connectivity Protocols
Robotics & Navigation	Autonomous Navigation, LiDAR-Based Obstacle Avoidance, 6-DOF Motion Control
VLSI / Digital Design	RTL-to-GDSII Flow, Digital SoC Design & Planning (Open-Source EDA Tools)
Professional	Project Management, Technical Leadership, Cross-Functional Teamwork, Technical Communication

Experience

Instruments Universal – Vadodara, India

Industrial Intern

2026

- Completed a 5-month industrial internship gaining applied exposure to instrumentation and industrial electronics workflows.

Monk9 Tech Private Limited – Rajkot, India

VLSI Design Intern

2024 – 2025

- Executed digital SoC design and planning using open-source EDA tools, covering floorplanning and design-stage verification.
- Implemented a complete RTL-to-GDSII physical design flow using open-source tools, strengthening foundations in digital IC design.

Key Projects

Autonomous Underwater Vehicle (AUV) – Robofest 4.0 & SAUVC 2025

Team Lead

2024 – 2025

- Designed and built a cost-effective underwater robot with LoRa-based wireless communication, real-time surveillance, and 6-DOF maneuverability; secured a design patent and a utility patent in India for its innovations.
- Achieved 6th place among international teams at SAUVC 2025, Singapore, and Runner-Up at the Robofest 4.0 National Competition.

Autonomous Kiwi Bot

Independent Project

2025

- Developed an autonomous mobile robot using LiDAR-based navigation and real-time obstacle avoidance algorithms; co-authored a technical paper documenting the methodology.

Smart Three-Phase Motor Automation System

Final-Semester Industrial Defined Project

2025 – 2026

- Engineered a low-cost automation system for three-phase motors with IoT-based real-time monitoring, fault detection, and remote control for agricultural and industrial use.
- Won the Project Expo for demonstrated practical impact and system reliability.

Additional Projects: Smart RC Car, Digital Clock without Microcontroller, Smart Agri Starter, Smart IoT-Based Dustbin, Fully Automatic Water Level Controller, Contactless Phase Wire Detector, Smart Door Lock System

Publications & Patents

- Paper:** Low-Cost Smart Three-Phase Motor Automation and Real-Time Monitoring Solution for Agriculture and Industry (2025–2026)
- Paper:** Autonomous KIWI Bot with LiDAR-Based Navigation and Obstacle Avoidance (2025)
- Design Patent (447676-001):** Granted in India for an underwater robot with novel ornamental and functional features.
- Utility Patent (Application No. 202521018561 A):** Granted in India for an underwater robot application enhancing marine exploration, inspection, and data collection.

Achievements & Leadership

- Awarded the **Gaurav Puraskar** and a **INR 10 Lakh (approx. USD 12,000) cash prize** for innovation in underwater robotics at Robofest 4.0 (2024–2025).
- Represented India at **SAUVC 2025**, Singapore – a global autonomous underwater vehicle competition.
- National Cadet Corps (NCC):** Earned the Cadet rank and passed the “B” Certificate examination (2022–2024).

Languages: Gujarati (Native) | Hindi (Fluent) | English (Professional Working Proficiency)